

```
import pandas as pd
import numpy as np
```

```
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
```

## Read Data

```
df=pd.read_csv('/content/retail_sales_dataset.csv')
```

## Head Of Data

```
df.head()
```

|   | Transaction ID | Date       | Customer ID | Gender | Age | Product Category | Quantity | Price per Unit | T Am |
|---|----------------|------------|-------------|--------|-----|------------------|----------|----------------|------|
| 0 | 1              | 2023-11-24 | CUST001     | Male   | 34  | Beauty           | 3        | 50             |      |
| 1 | 2              | 2023-02-27 | CUST002     | Female | 26  | Clothing         | 2        | 500            |      |
| 2 | 3              | 2023-01-13 | CUST003     | Male   | 50  | Electronics      | 1        | 30             |      |
| 3 | 4              | 2023-05-21 | CUST004     | Male   | 37  | Clothing         | 1        | 500            |      |
| 4 | 5              | 2023-~ ~ ~ | CUST005     | Male   | 30  | Beautv           | 2        | 50             |      |

Next steps:

[Generate code with df](#)[New interactive sheet](#)

## Check Null Value

```
df.isnull().sum()
```

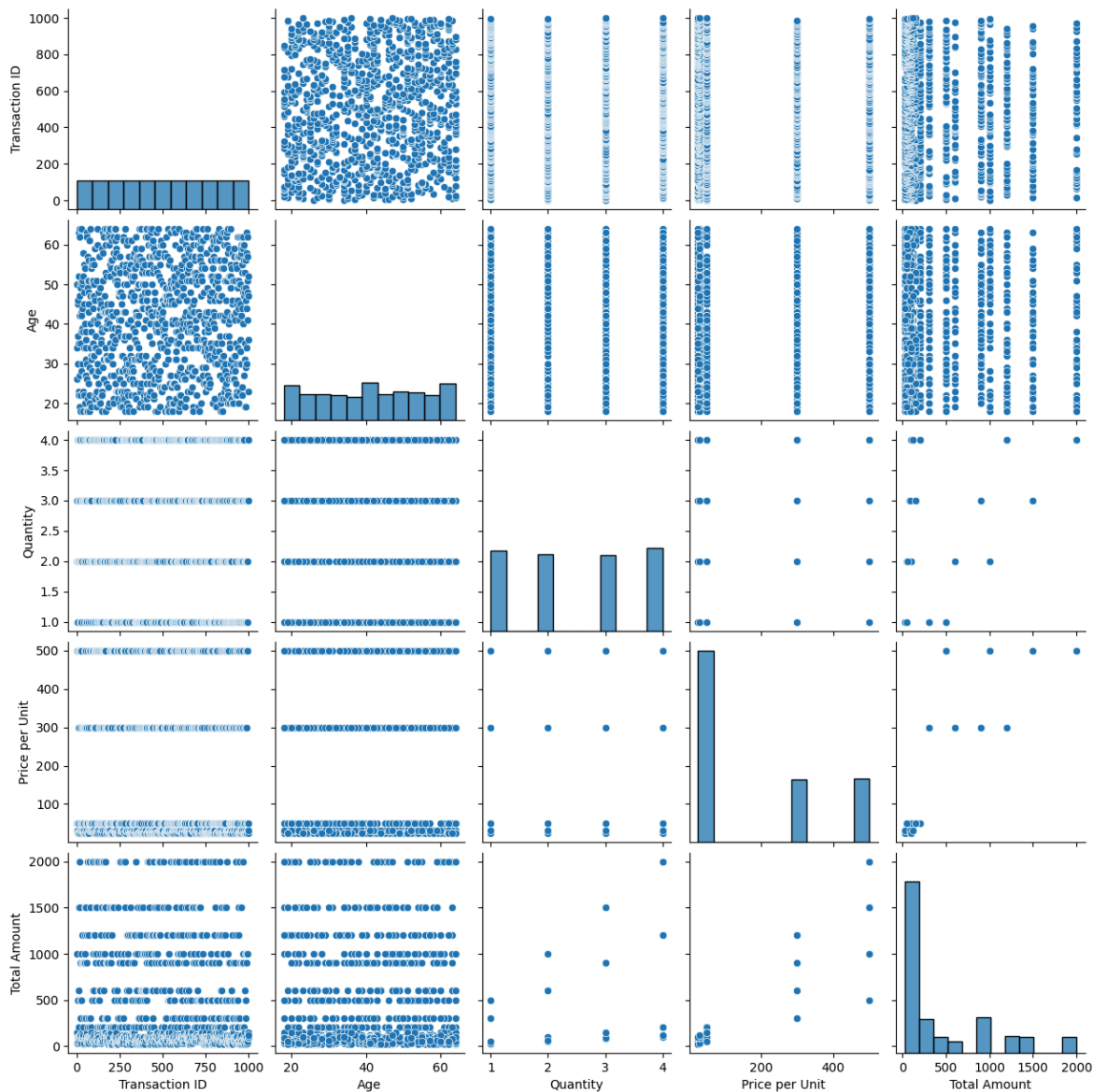
|                         | 0 |
|-------------------------|---|
| <b>Transaction ID</b>   | 0 |
| <b>Date</b>             | 0 |
| <b>Customer ID</b>      | 0 |
| <b>Gender</b>           | 0 |
| <b>Age</b>              | 0 |
| <b>Product Category</b> | 0 |
| <b>Quantity</b>         | 0 |
| <b>Price per Unit</b>   | 0 |
| <b>Total Amount</b>     | 0 |

**dtype:** int64

## PairPlot *italicised text*

```
sns.pairplot(data=df)
```

&lt;seaborn.axisgrid.PairGrid at 0x7e82ac344440&gt;



```
df['Date']=pd.to_datetime(df['Date'])
```

## Age Graph

```
#Pie Chart
```

```
#Gender Distribution  
df1 = df['Gender']  
fig = px.pie(df1,  
              names='Gender',  
              title='Gender Distribution')  
fig.show()
```

Gender Distribution

